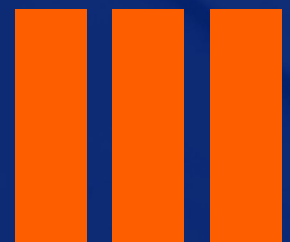




Scoring Method:

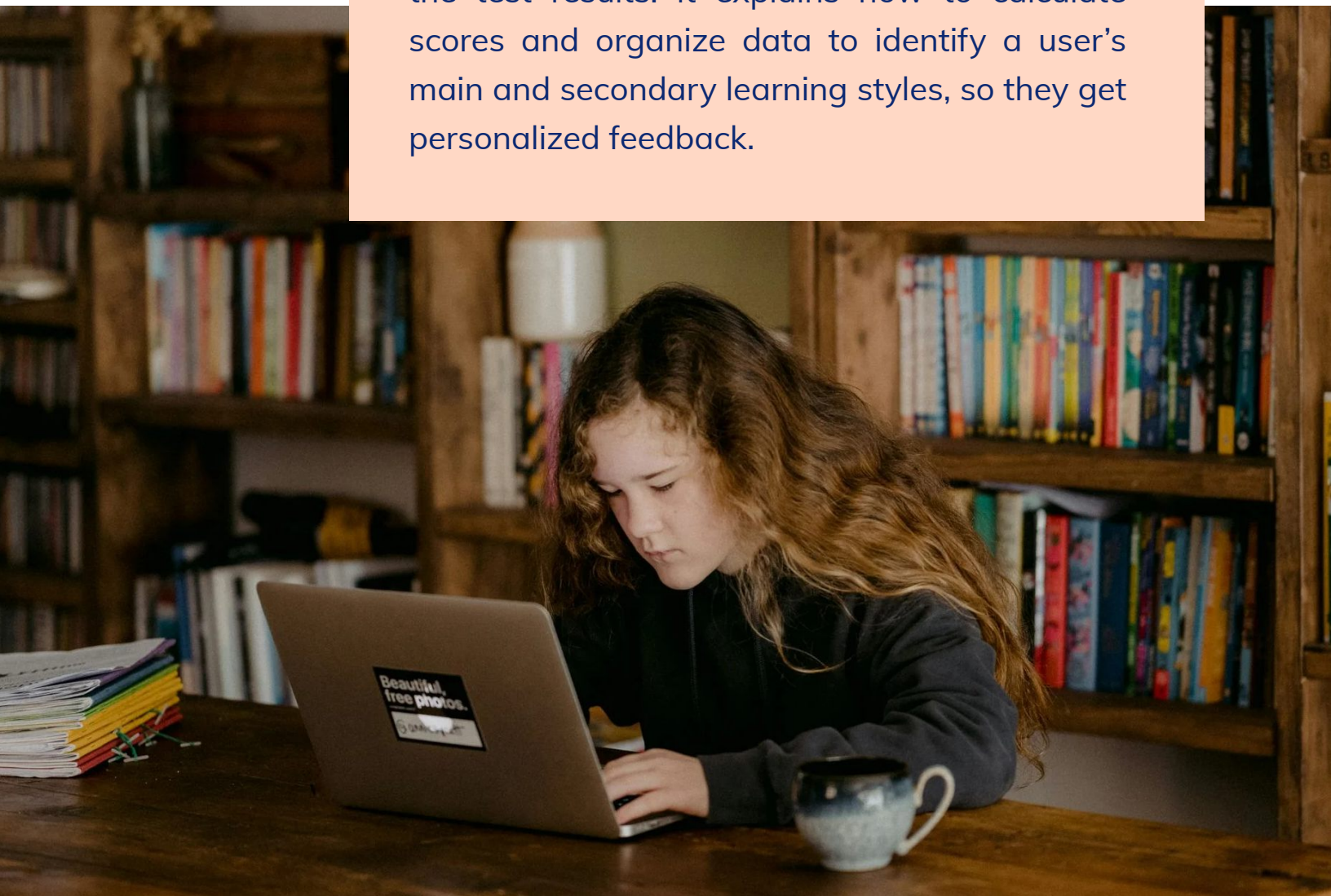
**Simple Backend
Instructions for
Report Generation**

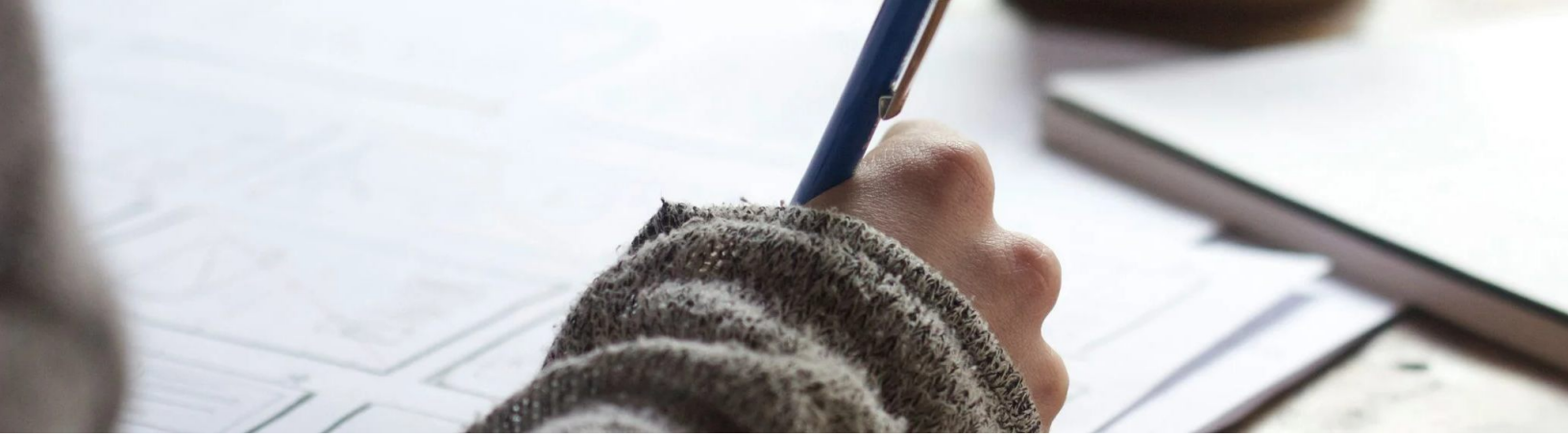


Scoring Method:

Simple Backend Instructions for Report Generation

This guide is for the backend development team to help create automatic reports based on the test results. It explains how to calculate scores and organize data to identify a user's main and secondary learning styles, so they get personalized feedback.





Scoring Steps for Backend Processing

1. Collect the Answers:

- Gather the user's answers for all 60 questions. Each answer will be one of these options: A, B, C, or D.
- Save these answers in an organized way, like a list or database, for easy processing.

2. Count the Choices:

- Count how many times the user picked each option:
 - **A_Count:** Number of times A (Visual) was chosen.
 - **B_Count:** Number of times B (Auditory) was chosen.
 - **C_Count:** Number of times C (Read/Write) was chosen.
 - **D_Count:** Number of times D (Kinesthetic) was chosen.
- Make sure the total adds up to 60 ($A_Count + B_Count + C_Count + D_Count = 60$).

3. Find the Main Learning Style:

- The option with the highest count is the user's **Dominant Learning Style**.
- If two or more options tie for the highest count, call it a **Mixed Style** (e.g., "Visual-Auditory Mix").

4. Check for a Secondary Style (if needed):

- If the highest count is 5 or more points above the next highest, focus only on the dominant style.
- If the second-highest count is less than 5 points below the dominant style, include it as a **Secondary Learning Style** (e.g., "Primary: Visual, Secondary: Kinesthetic").

5. Calculate Percentages for Charts:

- Turn the counts into percentages to show in graphs or charts:
 - Visual Percentage = $(A_Count / 60) * 100$
 - Auditory Percentage = $(B_Count / 60) * 100$
 - Read/Write Percentage = $(C_Count / 60) * 100$
 - Kinesthetic Percentage = $(D_Count / 60) * 100$
- Round the percentages to one decimal place (e.g., 33.3%).

6. Set Importance Levels:

- If a style has a count of 10 or less (under 16.7%), mark it as Low Preference in the report, meaning the user doesn't rely on it much.
- If a style has a count over 20 (above 33.3%), mark it as Strong Preference in the feedback to show it's a key strength.

How to Build the Reports

➔ Basic Report (Simple Summary for Users):

- Show the Dominant Style (or Mixed Styles if there's a tie).
- Include a bar chart or pie chart with the percentages for all four styles.
- Add a short description of the dominant style(s) and 3-5 study tips (from the detailed reports provided later).
- Mention a Secondary Style if it's within 5 points of the dominant one.

➔ Detailed Report (For Teachers or Counselors)

- Show the exact counts and percentages for all styles.
- Explain the Dominant and Secondary Styles using detailed information (from the detailed reports provided later).
- Point out "Low Preference" areas and suggest other ways to learn.
- Give specific advice for different subjects and note possible challenges based on the style(s).

➔ Storing Data for Analysis:

- Save user results in a database with these fields: User_ID, A_Count, B_Count, C_Count, D_Count, Dominant_Style, Secondary_Style, Timestamp.
- Allow analysis of patterns, like finding the most common learning styles in different age groups or classes.



Handling Errors

- If the total answers don't add up to 60 (e.g., some questions were skipped), mark the report as Incomplete and ask the user to finish the test.
- If all counts are equal and no dominant style stands out, label the user as a Balanced Learner and give tips for all styles.